Harsh Gupta

Roll Number: 22IM30014
B.Tech - M.Tech Integrated
Industrial and Systems Engineering
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Education

- **Indian Institute of Technology Kharagpur** ***2022 - 2027***
  Dual Degree, Industrial and Systems Engineering CGPA: 8.39/10

Experience

- **Practworks** ***Dec 2023 - Mar 2024***
  Machine Learning Engineer Intern
  – Constructed a comprehensive dataset and **concept dependency graph**, establishing inter-concept relationships
  – Developed a robust **synthetic data generation algorithm** (48,000 data points) to solve data scarcity problem
  – Led **testing** and **fine-tuning** using cross-validation techniques, achieving a model accuracy of **67%**
- **Ylytic** ***Sep 2023 - Nov 2023***
  Software Engineer Intern
  – Integrated **5 new data features** into the company's database, enhancing data-driven insights and decision-making
  – **Automated task execution** and implemented rigorous testing protocols, ensuring data processing and reliability
  – Optimized workflows, **reduced processing time(by 60%)**, and ensured error-free operations for **2.4 million data**

Projects

- **Azad Hall Website | Team Project** ***Oct 2023 - Feb 2024***
  Modernized Azad Hall's website with features and performance for seamless user interactions
  – Deployed **Django website on GCP** with Nginx, Gunicorn, and Docker for scalability and maintenance
  – Architected website to handle **concurrent users**, ensuring website responsiveness and performance
  – Integrated **PostgreSQL database** for efficient data management, enabling seamless website features.
- **Self-Evaluation of Academic Performance | Research Project** ***Jun 2023 - Oct 2023***
  Harnessed the power of smartphone usage data to build a novel model for predicting student GPA
  – **Preprocessed smartphone data**, cleaning and analyzing features to identify high-quality features
  – Evaluated regression models (**Lasso, Ridge, SVR**) using **K-fold cross-validation** for unbiased performance
  – Optimized predictive performance with Ridge regression, achieving a **Mean Absolute Error of 0.255**

Competitions

- **Voice and Image Controlled Inventory Management | Bhashini Sprint** ***Mar 2024 - May 2024***
  Build multi-modal inventory system with voice control, image tracking, and integration
  – Implemented an **audio recording system** with Bhashini ASR integration, enabling **voice-based inventory input**
  – **Fine-tuned** a small LLM for precise **text-to-command conversion**, enhancing accuracy and response time.
  – Designed and implemented MySQL database to support **efficient inventory management** via a custom algorithm

Skills

Languages: Python, C++, C, Javascript **Developer Tools**: Git (Version Control), PyCharm, Docker, DevOps, Postman, Figma **Frameworks and Libraries**: Flask, Scikit-learn, Hugging Face, NLTK, spaCy, Matplotlib, Pandas, Numpy, FastAPI **Cloud/Databases**: MongoDB, AWS, GCP

Relevant Coursework

Programming and Data Structures, Algorithm Lab, Operation Research lab, Probability and Statistics

Positions of Responsibility

– **Associate Quant Researcher** Quant Club, IIT Kharagpur ***May 2023 - Present***
  ∗ Implemented **algorithmic trading strategies** outlined in the White Paper and simulated trading environments.
  ∗ Fostered a collaborative learning environment within Quant Club, **promoting skill development and engagement**

Extra-Curricular Activities

– **Volunteer,** National Service Scheme ***Nov 2022 - Apr 2024***
  Design and execute engaging **workshops and outreach programs** to promote community involvement

Achievements

– **Bhashini Sprint** Winner of a government-organized competition, securing **first place nationwide**
– **JEE Advanced** Ranked in the top **5%** in JEE Advanced out of **250,000** candidates across India